Flutter Onset of a Fully Passive Flapping-Plate Energy Harvester

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Abstract

A flat plate mounted on linear and torsional springs in a steady stream can extract energy from the flow without any actuation, but only once the free-stream velocity exceeds the flutter-onset velocity and sustained limit-cycle oscillations develop. Published predictions of this onset velocity for the canonical two-degree-of-freedom plate disagree markedly at some frequency ratios; when the uncoupled heave and pitch frequencies are equal, two recent studies differ by a factor of four. This disagreement is examined here using a two-dimensional discrete-vortex method. The method is first validated against a pitch–plunge benchmark at a Reynolds number of 10 000, for which the predicted extreme lift coefficients (14 and −16) agree closely with experiment (14 and −15). It is then used to bracket the flutter-onset velocity of a plate of thickness 2.3 % of the chord, with structural parameters identical to those of an earlier study, at heave-to-pitch frequency ratios of 0.25, 0.5, 0.75 and 1. The reduced onset velocity is found to lie between 1.5 and 2.0, 1.3 and 1.4, 1.1 and 1.2, and 2.0 and 2.5 respectively. At a frequency ratio of unity the results support the higher of the two published values ($U_{\mathrm{F}}^{*} = 1.83$) and show no sign of instability near the lower ($U_{\mathrm{F}}^{*} = 0.44$). The supercritical responses are dominated by the formation and shedding of large leading-edge vortices, and at a frequency ratio of 0.75 the amplitude of the limit cycle is found, unexpectedly, to decrease as the velocity increases.

Keywords: flapping-foil energy harvester; fully passive; flutter onset; limit-cycle oscillation; discrete-vortex method; leading-edge vortex

1. Introduction

Tidal streams, ocean currents and rivers form an attractive part of the mix of low-carbon generation that future energy targets will require (MacKay, 2008). Because water is some thousand times denser than air, a current carries roughly a thousand times the power of a wind of the same speed, and the global hydrokinetic resource has been estimated at up to 3 TW (Young et al., 2014). Most tidal-stream devices are turbines with horizontal or vertical axes. These operate in well-understood flow regimes, but their rotating blades raise concerns about blockage by debris and harm to marine life (Xiao and Zhu, 2014).

Oscillating foils offer a bio-inspired alternative. Fish and insects propel themselves efficiently by oscillatory rather than rotary motion (Triantafyllou et al., 2004), and some fish recover energy from vortices shed upstream by their own dorsal and ventral fins (Zhu et al., 2002). Operated in reverse, a foil that heaves and pitches in a current extracts energy from the flow. In comparison with turbines, oscillating foils have low tip speeds and generate little noise, both of which reduce their environmental impact, and they combine low centrifugal stresses with a wide, shallow, rectangular swept area that suits shallow channels (Xiao and Zhu, 2014; Young et al., 2014).

Mechanically, the simplest such harvester is fully passive: a foil mounted on a linear (heave) spring and a torsional (pitch) spring, with power taken off by a damper. No actuators are required, but power is produced only once the free-stream velocity exceeds the flutter-onset velocity, at which sustained limit-cycle oscillations (LCOs) develop. Classical linear theory (Theodorsen, 1935; Theodorsen and Garrick, 1942) does not predict this onset accurately for thin aerofoils

at low Reynolds number, because the aerodynamics in this regime are dominated by nonlinear viscous effects (Lee et al., 1999; Amandolese et al., 2013), and in particular by the formation and shedding of leading-edge vortices (LEVs). High-fidelity solvers for the coupled fluid–structure problem capture these effects (Kamakoti and Shyy, 2004), but are costly when a large space of parameters is to be explored.

For the canonical flat plate with two degrees of freedom, the published onset velocities disagree. At low ratios $\bar{\omega}$ of the uncoupled heave and pitch frequencies, Murua et al. (2010) and Wang et al. (2017) obtain similar values, but at $\bar{\omega} = 1$ they report $U_{\mathrm{F}}^{*} = 0.44$ and $U_{\mathrm{F}}^{*} = 1.83$ respectively, values that differ by a factor of four. The present paper uses a discrete-vortex method, DIVEX (Taylor and Vezza, 1999), whose low computational cost permits a systematic search, to examine this disagreement. The parameters that govern the performance of oscillating-foil harvesters are reviewed in § 2, and the method and aeroelastic model are described in § 3. The method is validated against the pitch–plunge benchmark of Ramesh et al. (2014) in § 4. The flutter-onset velocity is then bracketed at four frequency ratios, using exactly the structural parameters of Wang et al. (2017) (§ 5), and the implications of the results are discussed in § 6.

2. Background

2.1. Harvesting and propulsion

Although flapping-foil propulsion and energy harvesting appear similar, they differ fundamentally (Xiao and Zhu, 2014). A propulsor transfers energy from the foil to the fluid and leaves a reverse (thrust-producing) von Kármán wake; a harvester extracts energy from the fluid and leaves a conventional, drag-producing wake. A propulsor is judged by the thrust it produces in the direction of travel, a harvester by the magnitude and relative phase of the lift and the pitching moment. Aeroelastic flutter, an instability to be avoided in aircraft, bridges and cables, is the operating mechanism of a passive harvester, the aim of design being to reach sustained, high-frequency LCOs at a predictable velocity. The aerodynamics in this regime are strongly nonlinear. At high reduced frequency and low Reynolds number, LEVs make a major contribution to the lift, as they do in insect flight (Dickinson and Götz, 1993; Ellington et al., 1996; Ellington, 1999; Shyy and Liu, 2007), rather than acting as the precursor to damaging dynamic stall familiar from helicopter and wind-turbine aerodynamics (Leishman, 2006).

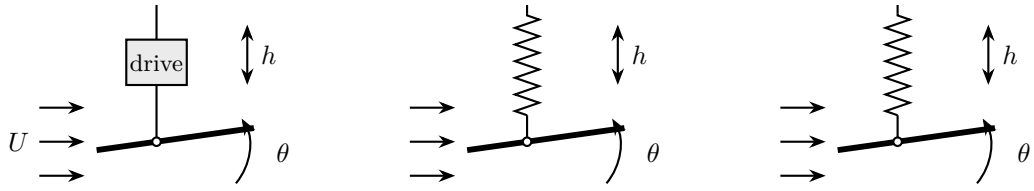
2.2. Modes of activation

Three kinematic configurations, or modes of activation, have been studied (figure 1) (Xiao and Zhu, 2014). In the fully prescribed mode both pitch and heave are driven, and the net power follows from the time-averaged balance between the power supplied and that extracted; this mode is simple to analyse and has received most attention, but is largely hypothetical. In the semi-passive mode the pitch is driven by a motor and the foil responds with a passive heave from which power is taken by a damper. In the fully passive mode both motions are governed by springs, and no input of power is needed beyond, possibly, an initial perturbation. It is this last mode that is considered here.

2.3. Governing parameters

The performance of a flapping-foil harvester is sensitive to many parameters, of which those most relevant here are summarised below.

With the other parameters fixed, the efficiency of extraction is highly sensitive to the frequency of oscillation, rising to a peak and falling sharply beyond it; the frequency also alters the wake, which matters for arrays of devices (Xiao and Zhu, 2014). The coefficient of power is maximised



heave: prescribed
pitch: prescribed

(a) fully prescribed

heave: passive
pitch: prescribed

(b) semi-passive

heave: passive
pitch: passive

(c) fully passive

Figure 1. Modes of activation of an oscillating-foil harvester in a free stream U: (*a*) fully prescribed; (*b*) semi-passive; (*c*) fully passive. After Xiao and Zhu (2014).

by a large heave amplitude at low frequency (Davids, 1999; Lindsey, 2002; Dumas and Kinsey, 2006; Xiao et al., 2012), whereas efficiency favours a small heave amplitude at high frequency (Young et al., 2014). Extraction is greatest when pitch and heave are about 90° out of phase (Davids, 1999). Since moving the pivot is equivalent to changing this phase, pivots are usually placed between the quarter- and third-chord points, near the centre of pressure; efficiency is high for pivots ahead of mid-chord, and for phases above 90° the optimum pivot moves forward (Davids, 1999; Xiao and Zhu, 2014).

Separation at the leading edge produces the LEV, so that a larger angle of attack produces a larger and stronger vortex and, transiently, more lift. Figure 2 illustrates this for an impulsively started static plate at $Re = 10\,000$ and angles of attack of 10° and 45°.

Efficiency increases modestly with Reynolds number, from 32.7 % at $Re = 500$ to 36.4 % at $Re = 10\,000$ in one study, an increase attributed to reduced viscous diffusion, which thins the effective foil and increases the forces (Xiao and Zhu, 2014). Bio-inspired corrugated profiles have been reported to improve the efficiency by about 6 % relative to a NACA 0012 aerofoil (Le et al., 2013), and prescribed flexibility can raise the peak lift and shift the phase between force and velocity favourably (Liu et al., 2013), although excessive flexibility degrades the forces (Young et al., 2014). Experiments on finite spans at $Re \approx 500\,000$ produced less power than equivalent two-dimensional models, because strong vortex shedding is three-dimensional (Simpson et al., 2008a,b); at the Reynolds number considered here, fifty times lower, the vortices are correspondingly weaker. Finally, the Betz limit caps the fraction of the power flux through the swept area that a single rotor can extract at $16/27 \approx 0.593$; arrays of oscillating foils, in which downstream foils exploit the wakes of those upstream, have been argued to exceed it (Kinsey and Dumas, 2012; Vennell, 2013), but for an isolated harvester the classical limit remains the appropriate reference. To isolate the fundamental aeroelastic behaviour, a rigid flat plate is considered in two dimensions at $Re = 10\,000$.

3. Method

3.1. The discrete-vortex method

DIVEX (Taylor and Vezza, 1999) solves the two-dimensional incompressible Navier–Stokes equations in vorticity form by discretising the vorticity field into vortex particles. The body is represented by nodes joined into panels to form a polygonal surface, and a condition of zero mass flux through each panel determines the strength of the surface vorticity at each node. These strengths represent the entire vorticity distribution on the surface, and newly created vorticity is released into the flow as particles whose velocities follow from the Biot–Savart law. The body is assumed to be perfectly rigid, and its reference frame is shown in figure 3. Because the method resolves only the region of the flow that carries vorticity, its cost is much lower than that of

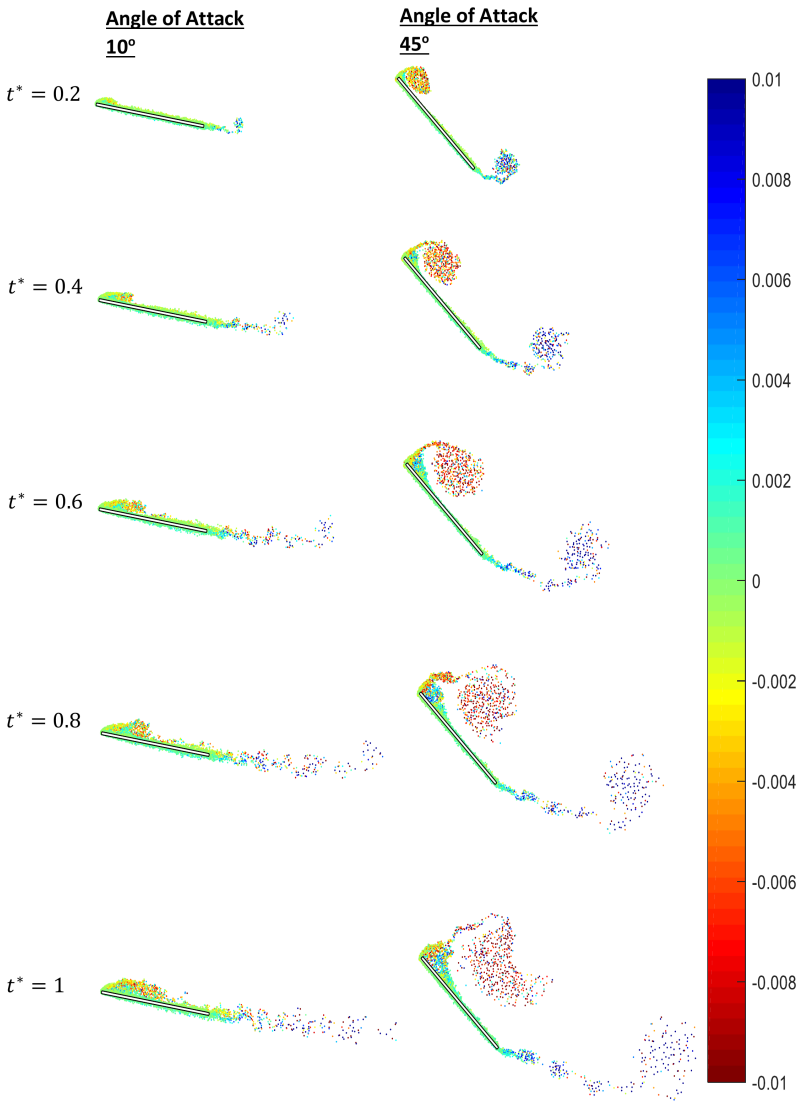


Figure 2. Vorticity field computed with DIVEX around an impulsively started static flat plate at $Re = 10\,000$, with the flow from left to right, at angles of attack of 10° (left) and 45° (right), for $t^* = 0.2$ to 1. Clockwise vorticity is negative and anticlockwise vorticity positive; all quantities are dimensionless.

grid-based computational fluid dynamics (CFD), which makes it well suited to the parameter sweeps required here.

3.2. Aeroelastic model

The fully passive system (figure 4) is a rigid flat plate of chord c and thickness $0.023c$, free to heave, with displacement h, and to pitch, through angle α, about a pivot a distance x_p aft of the leading edge. It is restrained by a linear spring of stiffness K_h and a torsional spring of stiffness K_α, and its centre of mass lies a distance x_α aft of the pivot. With h positive downwards and α positive nose-up, the motion is governed by the equations of the typical section (Theodorsen, 1935; Lee et al., 1999),

$$m\ddot{h} + mx_\alpha\ddot{\alpha} + K_h h = -L(t), \tag{3.1}$$

$$mx_\alpha\ddot{h} + I_\alpha\ddot{\alpha} + K_\alpha\alpha = M(t), \tag{3.2}$$

where m is the mass per unit span, I_α the moment of inertia about the pivot, and L and M the lift and the pitching moment about the pivot, computed by DIVEX at each time step. The

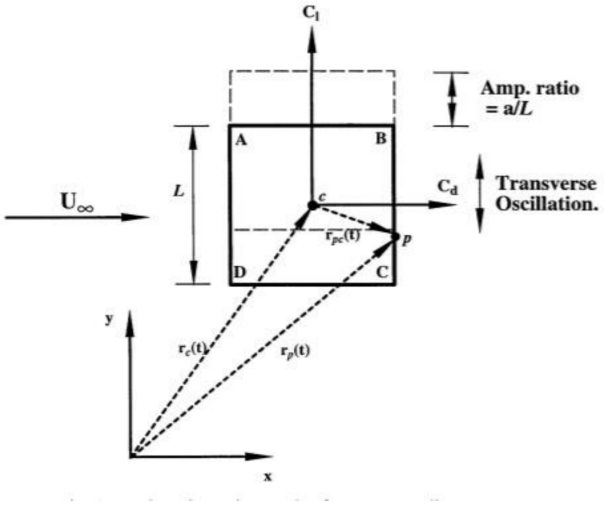


Figure 3. Orientation of the body and reference coordinate system used in DIVEX.

Table 1. Base parameters of the fully passive system, identical to those of Wang et al. (2017).

Parameter	Symbol	Value
Pivot location aft of leading edge	x_p	$0.35c$
Static unbalance	x_α	$0.05c$
Radius of gyration	r_α	0.5
Inverse mass ratio	μ^{-1}	0.05
Plate thickness	–	$0.023c$
Chord	c	0.0762 m
Free-stream velocity	U	$0.1312\,\mathrm{m\,s^{-1}}$
Reynolds number	Re	10 000

uncoupled natural frequencies are $\omega_h = (K_h/m)^{1/2}$ and $\omega_\alpha = (K_\alpha/I_\alpha)^{1/2}$, and the response is characterised by the frequency ratio and the reduced velocity,

$$\bar{\omega} = \frac{\omega_h}{\omega_\alpha}, \qquad U^* = \frac{U}{c\,\omega_\alpha}, \tag{3.3}$$

where U is the free-stream velocity. The base parameters, given in table 1, are identical to those of Wang et al. (2017).

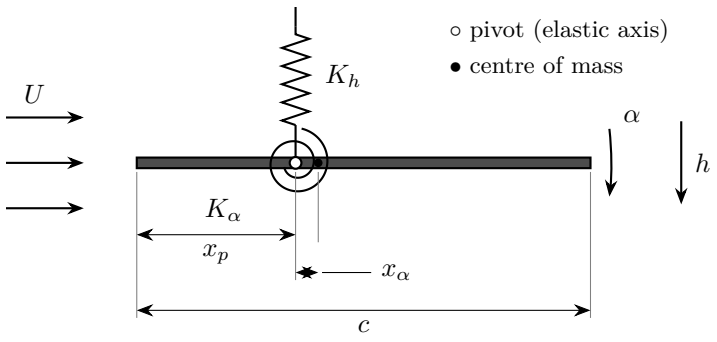


Figure 4. The fully passive flat plate with two degrees of freedom, showing the heave stiffness K_h, the pitch stiffness K_α, the pivot at x_p and the centre of mass a distance x_α aft of the pivot. After Wang et al. (2017).

Table 2. Parameters of benchmark case 3A of Ramesh et al. (2014).

Parameter	Value
Motion	sinusoidal pitch and plunge
Plunge amplitude, $h_{\max}/c$	0.05
Reduced frequency, $k_\alpha = k_h$	3.93
Phase, ϕ	90°
Pivot location from leading edge	$0.75c$
Time step, Δt^*	0.005
Number of time steps	8000 ($t^* = 40$)
Reynolds number	10 000

3.3. Bracketing the onset of flutter

The free-stream velocity and all geometric and inertial parameters were held fixed (table 1), and the reduced velocity was varied through the pitch natural frequency alone. From (3.3), with $\omega_\alpha = 2\pi f_\alpha$,

$$f_\alpha = \frac{U}{2\pi\, U^* c}, \qquad f_h = \bar{\omega} f_\alpha, \tag{3.4}$$

so that each pair $(\bar{\omega}, U^*)$ corresponds to a pair of input frequencies (f_h, f_α). For each frequency ratio, simulations were run over a range of U^*, and each response was classified from its pitch history and phase-plane trajectory as subcritical, if it remained of small amplitude and showed only flow-induced vibration; near-critical, if it showed large but irregular excursions without a closed limit cycle; or supercritical, if its amplitude grew and then saturated in a sustained LCO. The onset velocity U^*_{F} is then bracketed by the largest subcritical and smallest supercritical values of U^*. The simulations were run to a dimensionless time $t^* = tU/c$ of 150–300 or more, as indicated for each case.

4. Validation

The method was first validated against case 3A of Ramesh et al. (2014), who introduced a discrete-vortex method in which the shedding of LEVs is governed by a critical value of a leading-edge suction parameter (the LESP-modulated discrete-vortex method, LDVM). Ramesh et al. (2014) compared their method with water-tunnel experiments, with the theory of Theodorsen (1935) and with CFD computed with the CFL3D code (Krist et al., 1998; McGowan et al., 2011). The kinematics are given in table 2.

The reference results are reproduced in figure 5. The four sources differ noticeably: CFL3D gives the widest range of lift; LDVM follows the experiment closely, except for $t/T \approx 0.25$–0.6, where it falls below it and approaches CFL3D; and the theory of Theodorsen predicts a pure sinusoid, missing the plateau, or kink, near $t/T \approx 0.25$–0.7 that appears in the other results, because it cannot represent the formation of an LEV. The experiment is taken as the reference, the spread between the computational methods giving a measure of the accuracy that can be expected.

The results of DIVEX are shown over the whole run and over a narrower window in figure 6, and over a single period in figure 7. After an initial transient the response is periodic. The peak lift coefficient, $C_{\mathrm{l}} \approx 14$, matches the experiment, and the first kink, which begins near $C_{\mathrm{l}} \approx -10$, is reproduced; thereafter the cycle follows the experiment more closely than either LDVM or CFL3D, although a second kink present in the other results is not captured. The extreme values in table 3 show that DIVEX agrees with the experiment to within one unit of C_{l}, more closely than either of the other computational methods, from which it is concluded that the method is suitable for the flows dominated by LEVs that are considered below.

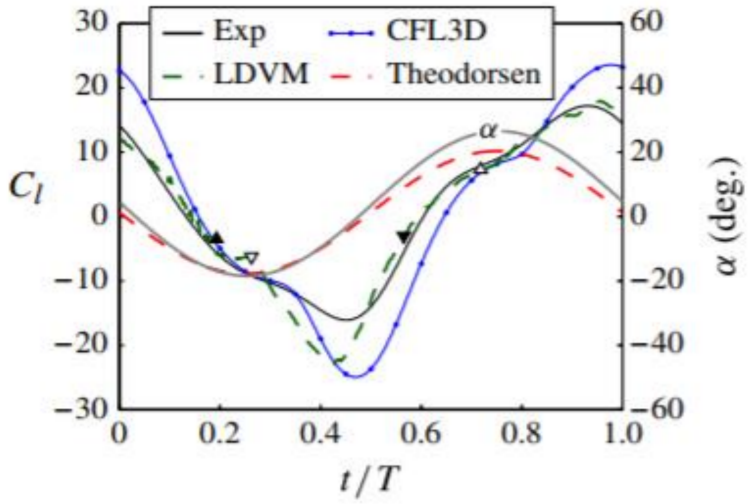


Figure 5. Lift coefficient over one cycle for case 3A: LDVM compared with experiment, CFD (CFL3D) and the theory of Theodorsen. Reproduced from Ramesh et al. (2014).

Table 3. Extreme lift coefficients for case 3A. Values other than those of DIVEX are from Ramesh et al. (2014).

	LDVM	CFL3D	Theodorsen	Experiment	DIVEX
$C_{\mathrm{l,max}}$	17	23	10	14	14
$C_{\mathrm{l,min}}$	−22	−25	−10	−15	−16

5. Onset of flutter

5.1. Frequency ratio of 0.25

Figure 8 shows the pitch response and phase-plane trajectories for $U^* = 1$, 1.5 and 2, computed to $t^* = 150$. At $U^* = 1$ the response does not grow and is clearly subcritical, and at $U^* = 1.5$ only small flow-induced fluctuations appear (figure 9*b*). At $U^* = 2$ the amplitude grows to an apparently steady LCO of about $\pm 50°$, with histories of lift and drag similar to those reported by Wang et al. (2017) (figure 9*a*); animations show large LEVs forming before each sharp increase in heave displacement, and their strength grows with the amplitude of pitch. The onset therefore lies in $1.5 < U^*_{\mathrm{F}} \leq 2$, consistent with the values of 1.5 (Murua et al., 2010) and 1.625 (Wang et al., 2017). Since Wang et al. (2017) report an LCO of about $\pm 60°$ already at $U^* = 1.6$, longer runs might place the onset nearer the lower end of this bracket.

5.2. Frequency ratio of 0.5

At $U^* = 1.4$ the amplitude of pitch begins to grow sharply at about $t^* = 300$ (figure 10), whereas at $U^* = 1.3$ the response remains small and shows no growth over a shorter run. The onset therefore lies in $1.3 < U^*_{\mathrm{F}} \leq 1.4$, a bracket that encloses closely the value of 1.313 obtained by Wang et al. (2017) and lies below the value of 1.563 obtained by Murua et al. (2010).

5.3. Frequency ratio of 0.75

At this ratio the two published predictions begin to diverge, Murua et al. (2010) giving $U^*_{\mathrm{F}} = 1.0$ and Wang et al. (2017) $U^*_{\mathrm{F}} = 1.313$, and a narrower range was therefore examined (figure 11). The responses at $U^* = 1.0$ and 1.1 are subcritical, and that at $U^* = 1.2$ develops a sustained LCO (figures 12 and 13), so that the onset lies in $1.1 < U^*_{\mathrm{F}} \leq 1.2$. On this evidence the value of Murua

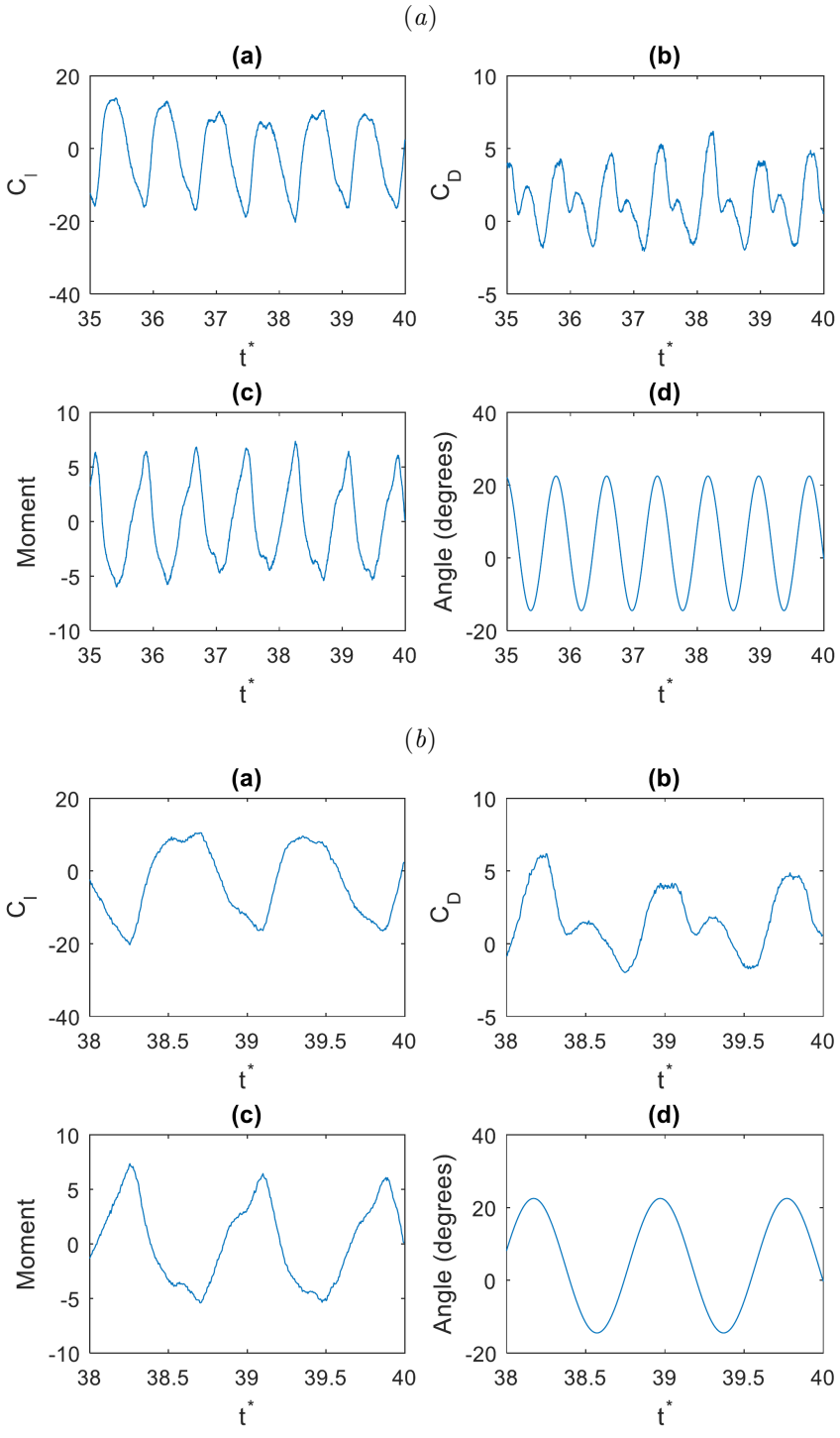


Figure 6. Results of DIVEX for case 3A over (a) the whole run, $0 \leq t^* \leq 40$, and (b) a narrower window of t^*. Each panel shows the lift, drag and moment coefficients and the pitch angle in degrees as functions of t^*.

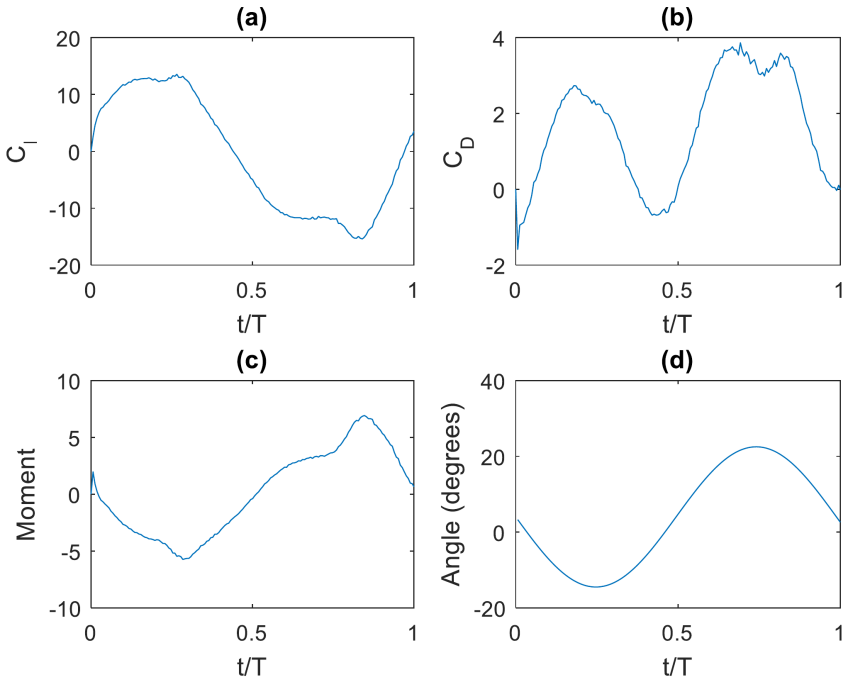


Figure 7. Results of DIVEX for case 3A over a single period: lift, drag and moment coefficients and pitch angle in degrees as functions of t/T.

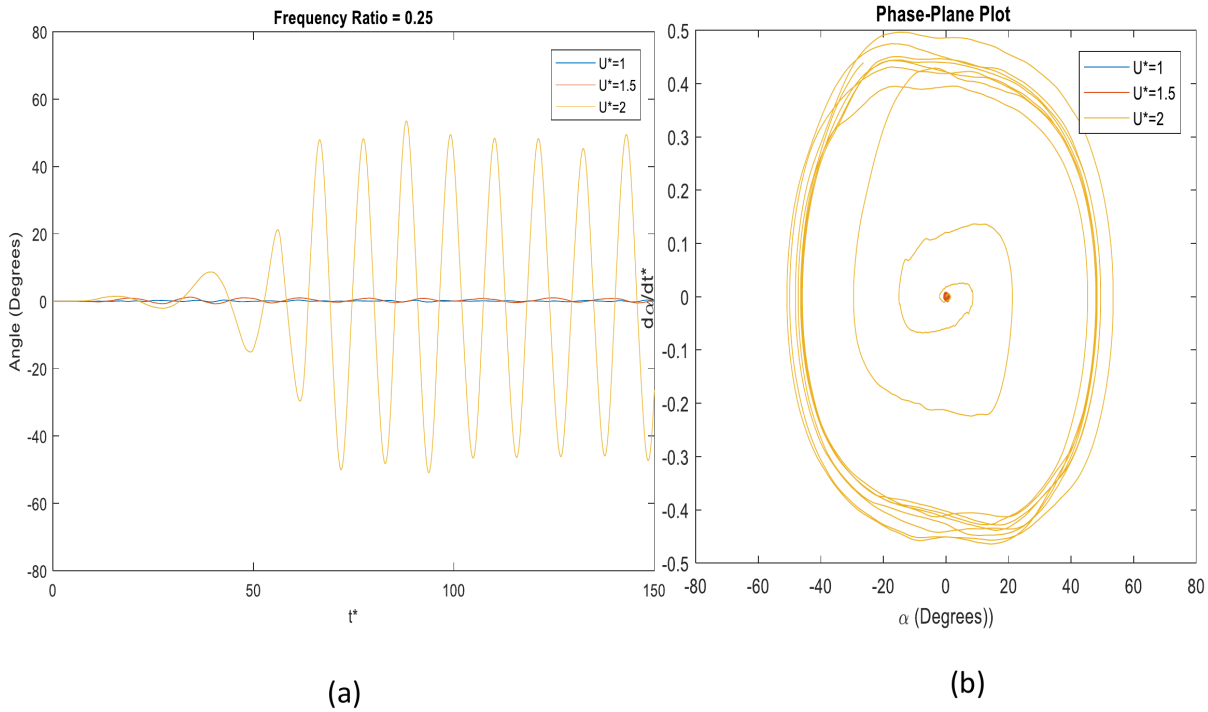


Figure 8. Frequency ratio $\bar{\omega} = 0.25$: (a) histories of pitch angle and (b) phase-plane trajectories for $U^* = 1$, 1.5 and 2.

et al. (2010) falls in the subcritical range and that of Wang et al. (2017) in the supercritical range.

As expected, flutter develops sooner as U^* is increased further. At $U^* = 1.45$, however, the resulting LCO has a smaller amplitude of pitch than at $U^* = 1.2$ (figure 14), whereas a similar or larger amplitude would normally be expected. This behaviour is considered further in § 6.

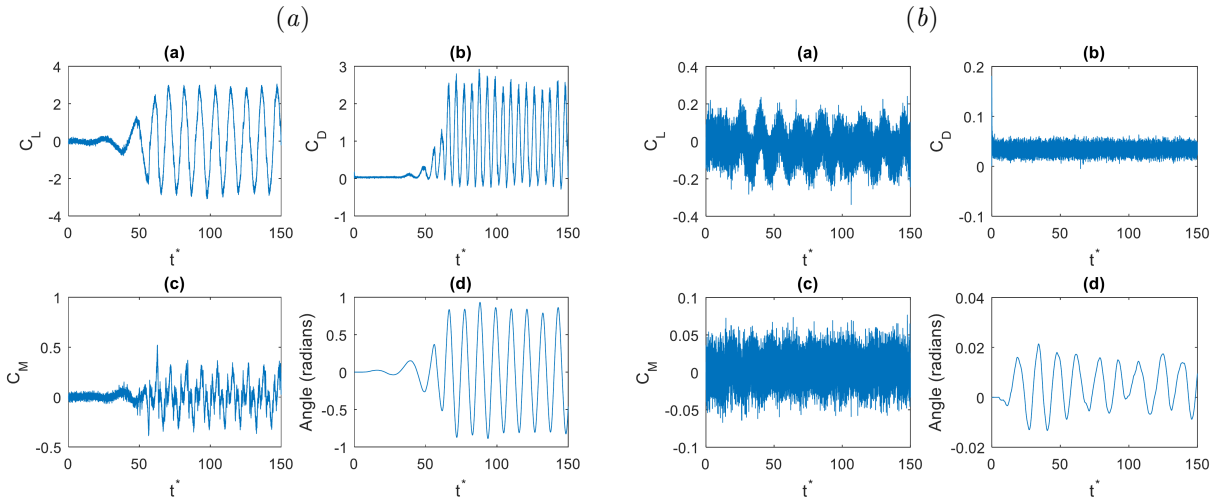


Figure 9. Frequency ratio $\bar{\omega} = 0.25$: lift, drag and moment coefficients and pitch angle in radians for (a) $U^* = 2$, which is supercritical, and (b) $U^* = 1.5$, which is subcritical.

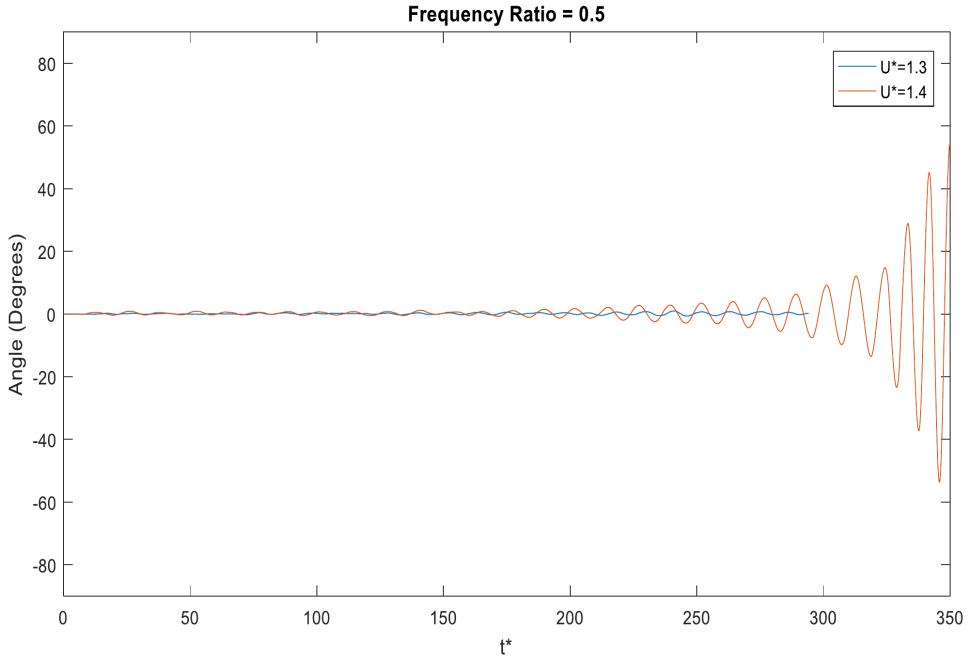


Figure 10. Frequency ratio $\bar{\omega} = 0.5$: pitch response for $U^* = 1.3$, computed over a shorter run that shows no growth, and for $U^* = 1.4$.

5.4. Frequency ratio of unity

It is at this ratio that the published predictions disagree most, Murua et al. (2010) giving $U^*_{\mathrm{F}} = 0.44$ and Wang et al. (2017) $U^*_{\mathrm{F}} = 1.83$. An initial wide sweep showed no sign of flutter anywhere near the lower value, and the search was therefore concentrated at higher values of U^*. At $U^* = 2$ the excursions in pitch are large but irregular and no closed limit cycle forms (figures 15 and 16), and this response is classified as near-critical. At $U^* = 2.5$ a sustained oscillation of large amplitude develops (figure 17). The onset therefore lies in $2.0 \lesssim U^*_{\mathrm{F}} < 2.5$, a result that supports the value of Wang et al. (2017) and is incompatible with that of Murua et al. (2010).

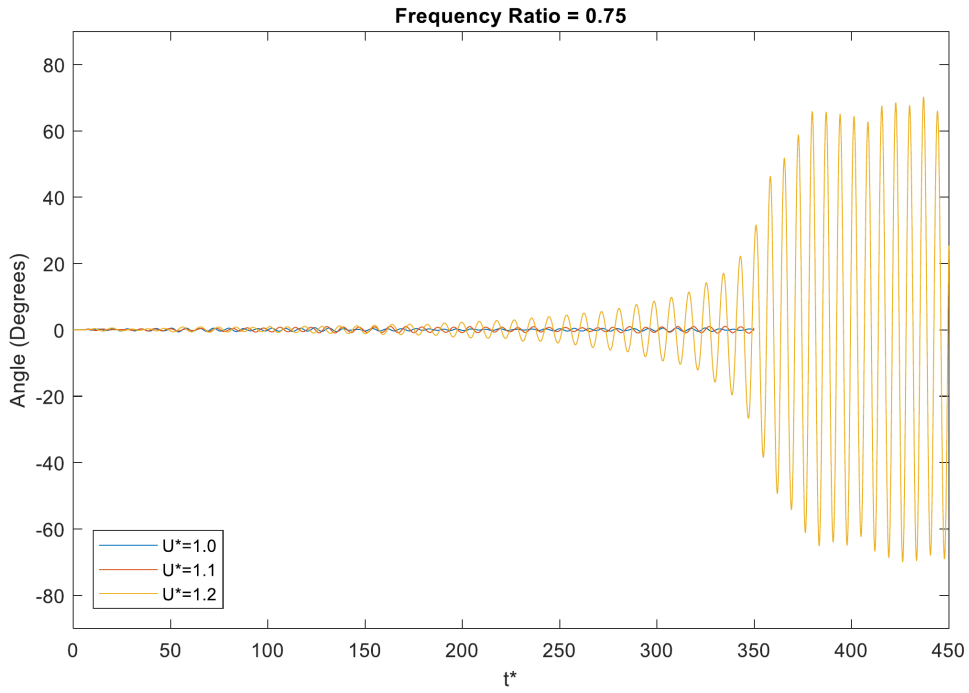


Figure 11. Frequency ratio $\bar{\omega} = 0.75$: pitch response for $U^* = 1.0$, 1.1 and 1.2.

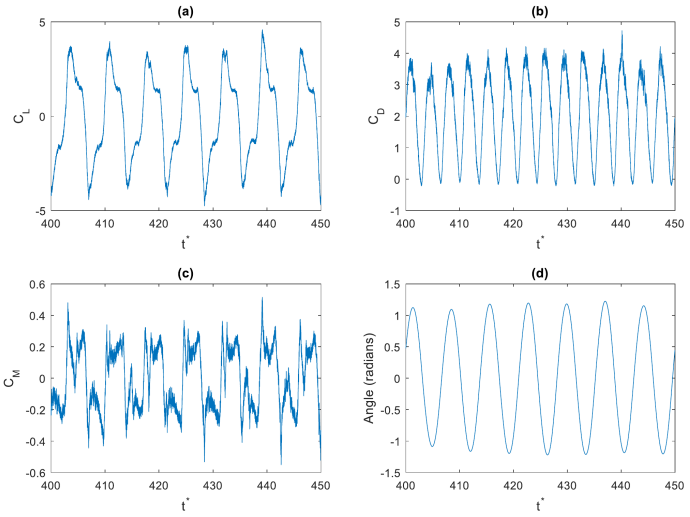


Figure 12. Frequency ratio $\bar{\omega} = 0.75$ and $U^* = 1.2$: lift, drag and moment coefficients and pitch angle in radians.

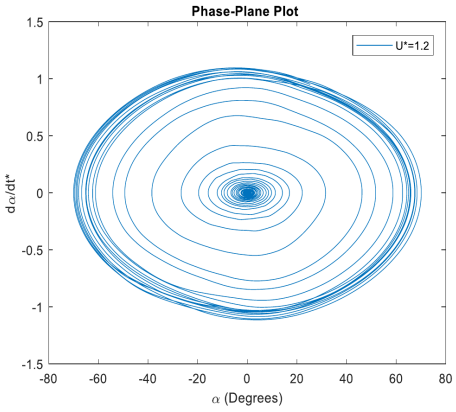


Figure 13. Frequency ratio $\bar{\omega} = 0.75$ and $U^* = 1.2$: phase-plane trajectory.

5.5. Summary of onset velocities

The onset velocities are summarised in table 4 and figure 18. At $\bar{\omega} = 0.25$ and 0.5 the three studies are broadly consistent. At $\bar{\omega} = 0.75$ the present bracket lies between the two published values, and at $\bar{\omega} = 1$ it lies close to the value of Wang et al. (2017) and far from that of Murua et al. (2010).

6. Discussion

The simulations show no amplification at reduced velocities near $U^*_F = 0.44$ for $\bar{\omega} = 1$, and place the onset at $U^* \approx 2$, slightly above the value of 1.83 obtained by Wang et al. (2017). Since DIVEX represents the nonlinear dynamics of the LEV directly, rather than through a linearised

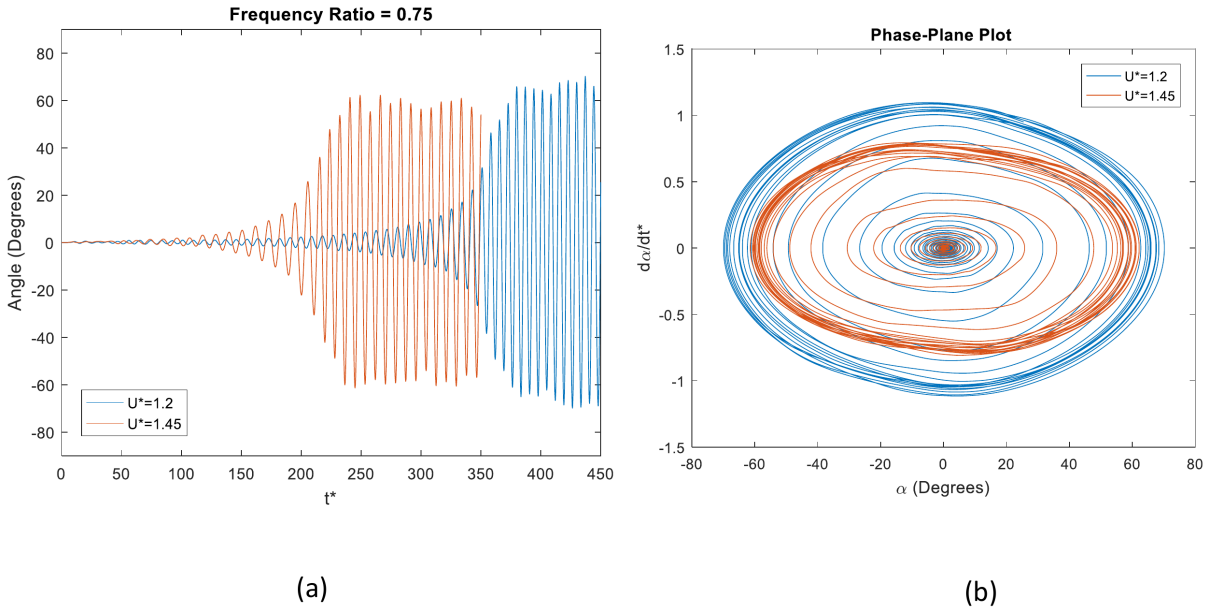


Figure 14. Frequency ratio $\bar{\omega} = 0.75$: (a) pitch response and (b) phase-plane trajectories for $U^* = 1.2$ and $U^* = 1.45$ (shorter run), both of which are supercritical. The LCO at $U^* = 1.45$ has the smaller amplitude.

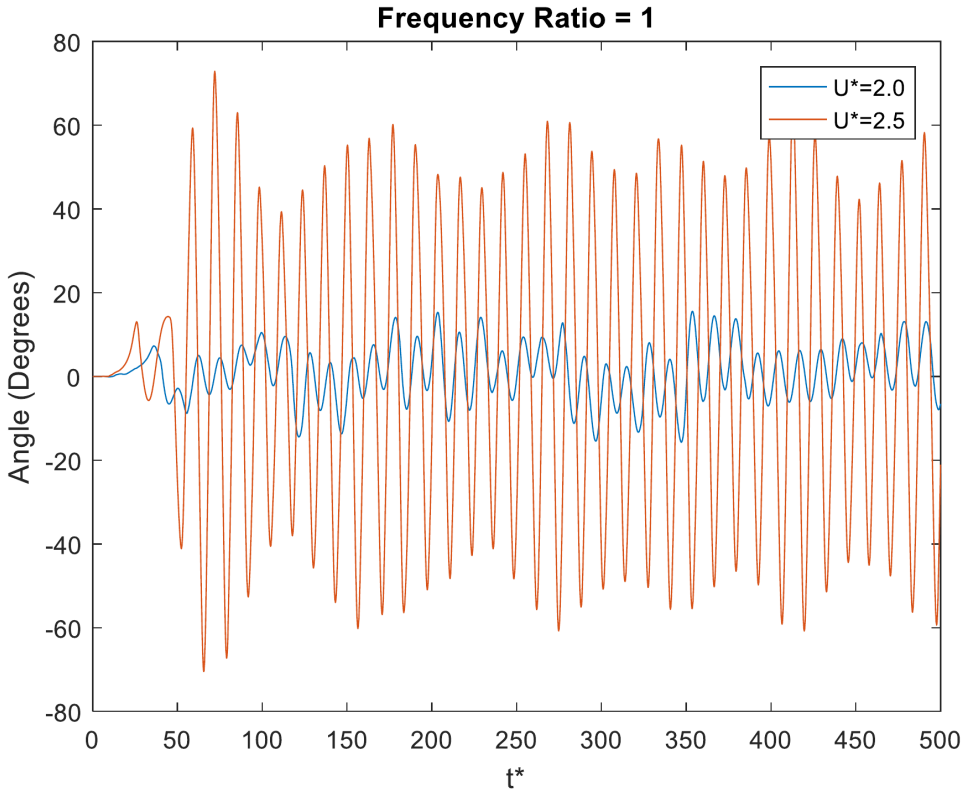


Figure 15. Frequency ratio $\bar{\omega} = 1$: pitch response for $U^* = 2$, which is near-critical, and $U^* = 2.5$, which is supercritical (shorter run).

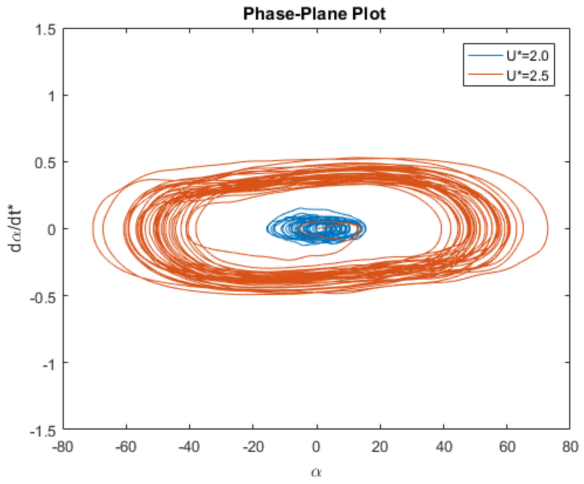


Figure 16. Frequency ratio $\bar{\omega} = 1$: phase-plane trajectories for $U^* = 2$ and 2.5.

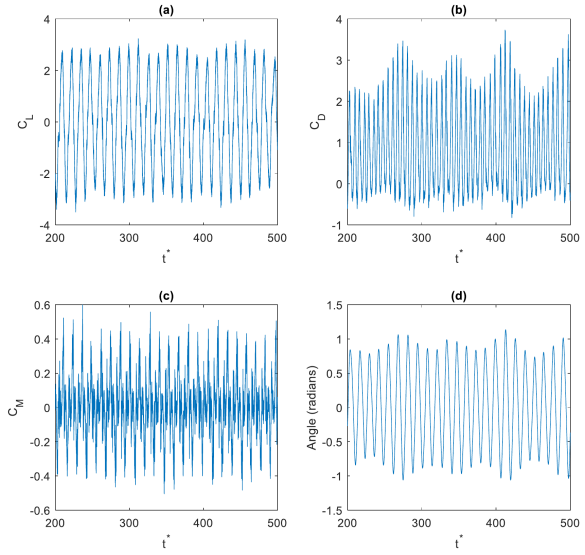


Figure 17. Frequency ratio $\bar{\omega} = 1$ and $U^* = 2.5$: lift, drag and moment coefficients and pitch angle in radians.

Table 4. Reduced flutter-onset velocity U^*_{F}. The present results are brackets between the largest subcritical and smallest supercritical values of U^*.

$\bar{\omega}$	Murua et al. (2010)	Wang et al. (2017)	Present study
0.25	1.5	1.625	1.5–2.0
0.50	1.563	1.313	1.3–1.4
0.75	1.0	1.313	1.1–1.2
1.00	0.44	1.83	2.0–2.5

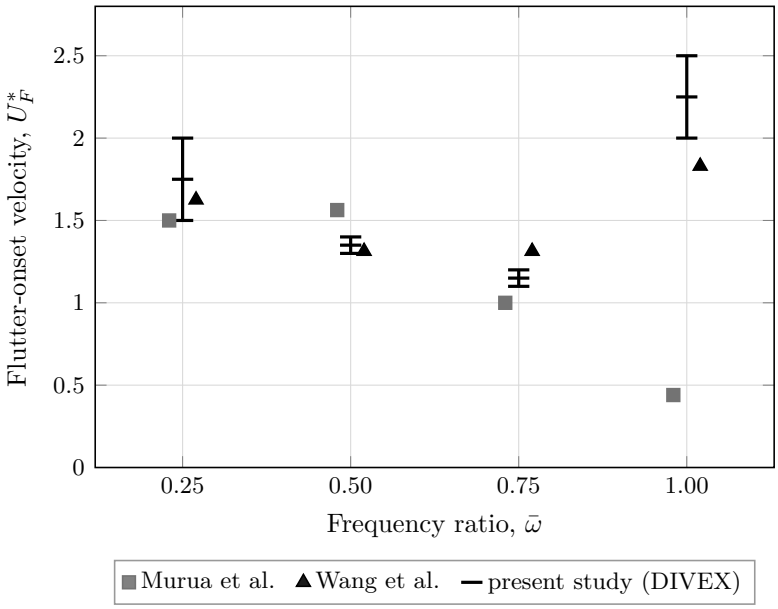


Figure 18. Reduced flutter-onset velocity as a function of frequency ratio. Bars show the brackets obtained in the present study; the published values are displaced slightly in $\bar{\omega}$ for clarity.

or attached-flow model, this agreement suggests that aerodynamics dominated by LEVs delay the onset of flutter when the heave and pitch frequencies coincide, a conclusion consistent with the failure of linear theory at low Reynolds number reported by Amandolese et al. (2013). The near-critical response at $U^* = 2$, with large, irregular excursions but no limit cycle, suggests moreover that onset is better described as a narrow transitional range than as a single critical value, below which the system is subcritical, within which its response is irregular and above which it settles into a sustained LCO. For the design of a harvester it is this range, rather than a single number, that matters.

In every supercritical case the rapid growth in the amplitude of pitch coincides with the formation and shedding of large LEVs, whose strength increases as the motion grows. The amplitude of the LCO is therefore set by a balance between aerodynamic forcing by the LEVs and the structural restoring moments, a balance that a linear model cannot capture. The smaller amplitude found at $U^* = 1.45$ than at $U^* = 1.2$ for $\bar{\omega} = 0.75$ remains unexplained. It may reflect a change in the timing of vortex shedding relative to the pitching motion, which would alter the exchange of energy per cycle, or it may be an artefact of the shorter run at $U^* = 1.45$, which may not have reached its final limit cycle; longer runs, together with an analysis of the vorticity field, are needed to distinguish between these explanations.

The brackets obtained here are coarse, of width between 0.05 and 0.5 in U^*, and the runs are of limited length, so that slowly growing responses near onset may have been classified as subcritical. The results would be sharpened by longer runs, by a small initial angle of pitch to shorten the transient, and by refining the brackets and adding further frequency ratios so as to obtain a continuous curve of onset velocity comparable with that of Murua et al. (2010). The analysis is restricted to a rigid flat plate in two dimensions at a single Reynolds number; its extension to finite spans, to flexible and profiled foils, and to the power extracted by a damper in the supercritical regime is the natural next step towards the design of practical harvesters.

7. Conclusions

A discrete-vortex method has been validated for pitching and plunging flows dominated by leading-edge vortices, and used to determine the onset of flutter of a fully passive flat-plate energy harvester with two degrees of freedom at $Re = 10\,000$. The method reproduces the experimental extreme lift coefficients of the benchmark of Ramesh et al. (2014), 14 and -15, to within one unit (14 and -16), more closely than either LDVM or CFL3D.

The reduced flutter-onset velocity lies between 1.5 and 2.0, 1.3 and 1.4, 1.1 and 1.2, and 2.0 and 2.5 at frequency ratios of 0.25, 0.5, 0.75 and 1 respectively. At a frequency ratio of unity, where published predictions differ by a factor of four, the present results support a value of about 1.8–2, as obtained by Wang et al. (2017), and show no sign of instability near the value of 0.44 obtained by Murua et al. (2010). The supercritical responses are dominated by large leading-edge vortices; onset is better described as a narrow transitional range than as a single critical velocity; and the amplitude of the limit cycle can decrease as the velocity increases.

These results show that an inexpensive vortex method can resolve conflicting aeroelastic predictions and identify the conditions under which a passive flapping-foil harvester will start of its own accord, a prerequisite for designing such devices to generate power.

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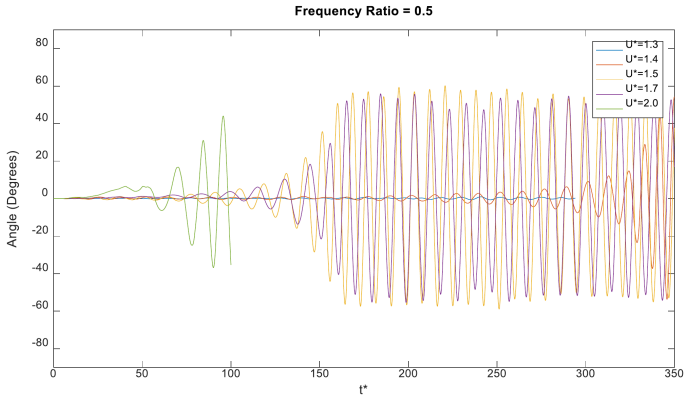
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Declaration of interests. The author reports no conflict of interest.

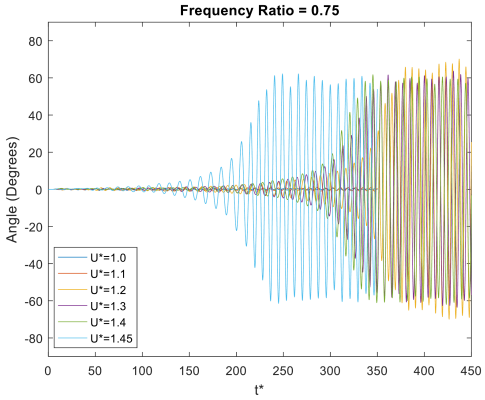
Data availability statement. The input parameters, time histories and animations of the simulations are available from the author on reasonable request.

A. Responses over the full range of reduced velocity

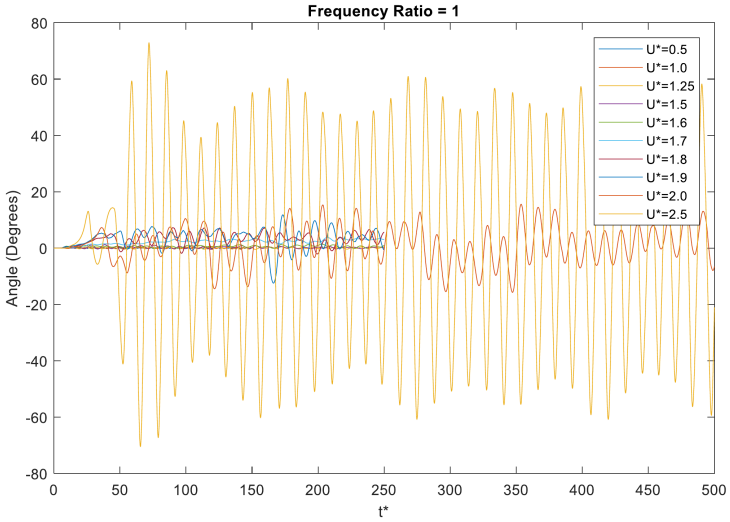
The full set of responses computed with DIVEX across the reduced velocities examined is shown in figure 19.



(a) $\bar{\omega} = 0.5$



(b) $\bar{\omega} = 0.75$



(c) $\bar{\omega} = 1$

Figure 19. Pitch responses computed with DIVEX over the full range of reduced velocity examined: (a) $\bar{\omega} = 0.5$; (b) $\bar{\omega} = 0.75$; (c) $\bar{\omega} = 1$.

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